

Day 9 [LSTM]

LSTM — Simple & Visual Explanation

Why Was LSTM Created? The Problem First.

Imagine you're reading this sentence:

Code



```
"The cat, which was sitting on the mat  
near the window in the old house, was hungry."
```

By the time you reach “**was hungry**”, you still remember “**the cat**” because your brain holds important things.

A **vanilla RNN forgets**. It's like having a whiteboard that gets partially erased at every word.

Code



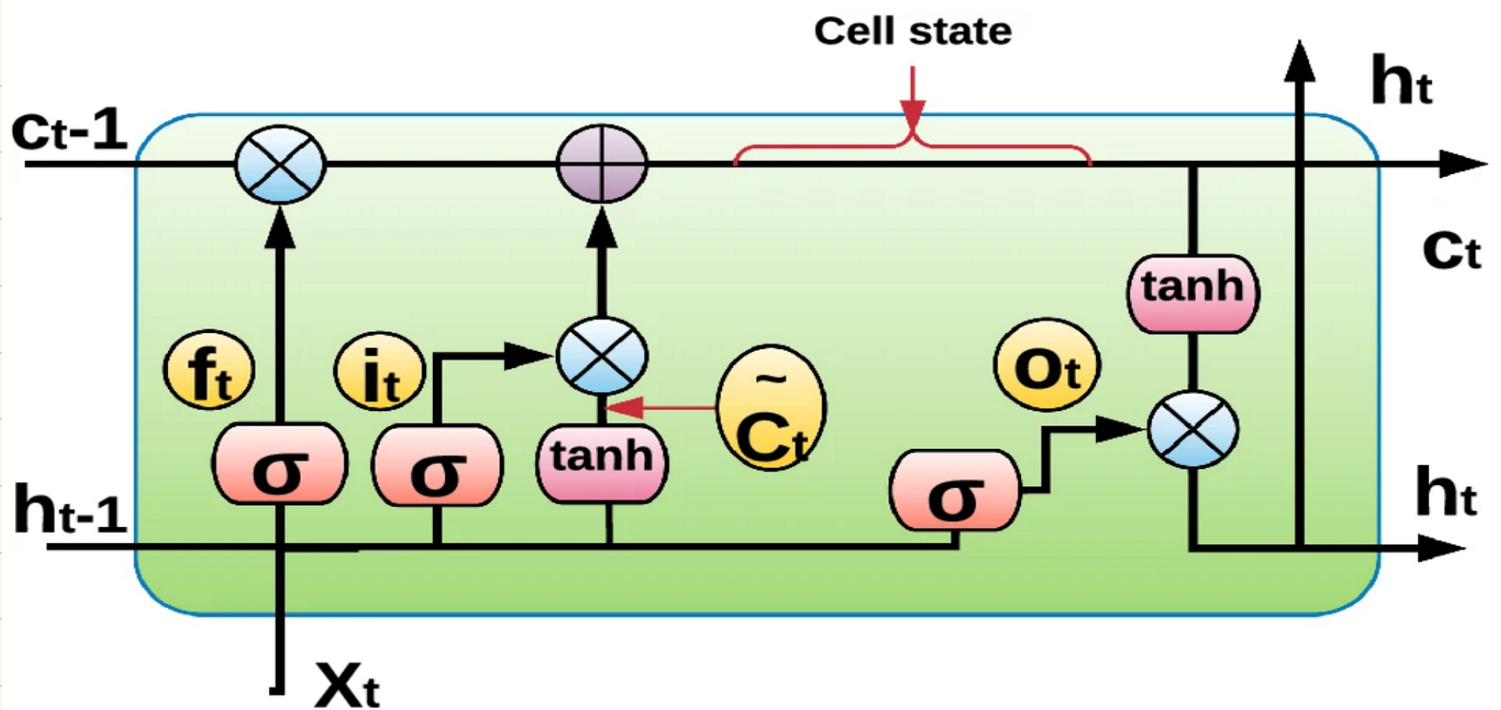
RNN Memory:

"The"	→	[the.....]	
"cat"	→	[.....cat.....]	
"which"	→	[.....which....]	← cat fading
"was"	→	[.....was.]	← cat almost gone
"sitting"	→	[.....sitting]	← cat GONE ❌

By the time we need "cat", it's erased.

Two core problems with RNN:

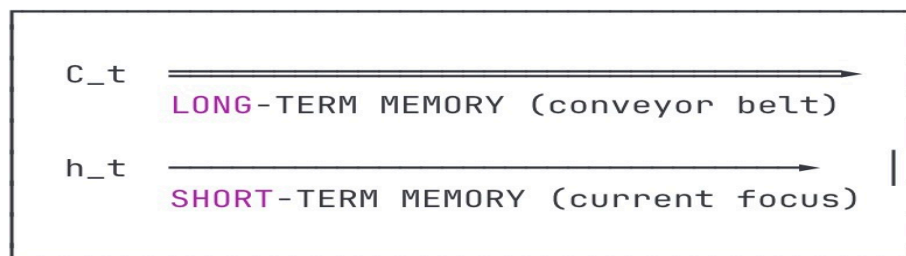
- Forgets early words (vanishing gradient)
- No control — everything gets overwritten blindly



LSTM's Big Idea — Two Memory Lanes

LSTM said: "Let's have two separate tracks?"

Code

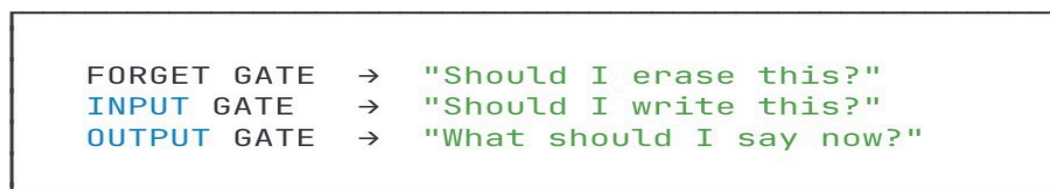


- $C_t \rightarrow$ like a **notebook** — carries important facts across long distances
- $h_t \rightarrow$ like **working memory** — what's relevant right this moment

The Three Gates — In Plain English

LSTM uses 3 gates (think of them as smart switches):

Code



Each gate outputs a value between 0 and 1:

Code

```
0 = fully closed (block everything)
1 = fully open   (let everything through)
0.5 = half open  (let some through)
```

Gate 1 — Forget Gate

“What old memory should I throw away?”

Code

```
Example sentence:
"He was a doctor. She is a ___"

When model sees "She" →
Forget gate says: erase gender info about "He"
Keep: sentence structure, context

Result: model correctly fills in female context
```

Code

```
Old Memory × Forget Gate

[ male=0.9, doctor=0.8, past=0.7 ]
  ×
[  0.0,      1.0,      0.8  ]
  =
[ male=0.0, doctor=0.8, past=0.56 ]
  ^^^^
  erased!
```

Gate 2 — Input Gate

“What new information should I add to memory?”

Code

```
Example:
"Aman went to the market"

When model sees "market" →
Input gate says: write location=market strongly
                  ignore irrelevant details

Memory now holds: person=Aman, location=market
```

Gate 3 — Output Gate

“What part of my memory is useful RIGHT NOW?”

Code

```
Example:
"The bank near the river flooded"

Cell memory holds:
[ bank=financial, bank=riverside, river=yes, flood=yes ]

Output gate looks at context → river + flood
→ Exposes only the RIVERSIDE meaning
→ Blocks the financial meaning

h_t = relevant output for this moment
```

Full LSTM Flow — Simple Diagram